

Assignment : Calculated Fields, Parameters & LOD Expressions

Question 1 :

What is a Calculated Field and why is it important in business dashboards?

Answer:

A Calculated Field is a new field created using formulas in Tableau to perform calculations on existing data. It helps users derive meaningful insights without changing the original dataset.

Importance in Business Dashboards:

- Creates custom metrics such as Profit Ratio, Growth Rate, and Average Sales.
- Automates calculations and reduces manual work.
- Helps in deeper analysis and better decision-making.
- Makes dashboards more interactive and informative.

Question 2 :

Why can't parameters change visuals directly in Tableau?

Answer:

Parameters are single-value inputs that store user selections, but they do not directly affect visualizations. They must be connected to a calculated field, filter, reference line, or other Tableau feature to influence the dashboard.

For example, a parameter can allow users to select a region, but a calculated field or filter is required to update the chart based on that selection.

Question 3 :

When should EXCLUDE LOD be used?

Answer:

EXCLUDE LOD (Level of Detail) is used when you want to calculate values while removing one or more dimensions from the current view.

Use Cases:

- Comparing individual values with overall totals.
- Calculating averages without considering a specific dimension.
- Analyzing performance against a broader benchmark.

Example:

If a dashboard shows sales by Sub-Category and you want the average sales at the Category level, EXCLUDE can remove Sub-Category from the calculation.

Question 4 :

Difference between FIXED and INCLUDE LOD expressions?

Answer:

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FIXED LOD

Calculates values at a specified level of detail regardless of dimensions in the view.

Independent of the visualization level.

Useful for creating benchmarks, totals, and KPIs.

Example: Total sales by region even when city is displayed.

INCLUDE LOD

Adds extra dimensions to the current view for calculation.

Depends on the dimensions present in the visualization.

Useful for detailed calculations within a view.

Example: Average customer sales while viewing regional sales.

Task 1: Delivery Performance Classification (Calculated Field)

Business Need

Operations wants to track delivery efficiency.

Task

Classify orders as Fast or Delayed.

Condition

Delivery $\leq$ 30 min $\rightarrow$ Fast

Else $\rightarrow$ Delayed

Answer :

# Sheet1 Unit Price	# Sheet1 Delivery Time Min	Abc Sheet1 Payment Mode	# Sheet1 Sales	# Sheet1 Cost	# Sheet1 Profit	Abc Calculation order classification
173	20	Cash	346	238.076	107.924	fast
128	37	UPI	512	312.717	199.283	delayed
167	55	Cash	334	215.354	118.646	delayed
135	41	UPI	135	94.237	40.763	delayed
106	14	UPI	212	177.432	34.568	fast
29	47	UPI	29	18.424	10.576	delayed

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Abc Category
 City
 -Abc order classification
 Abc Order Date
 # Order ID
 Abc Payment Mode
 Abc Product Name
 Abc *Measure Names*
 # Cost
 # Delivery Time Min

Task 2: Loss-Making Orders Identification

Business Need

High sales do not always mean high profit.

Task

Identify orders resulting in loss.

Answer :

×

```

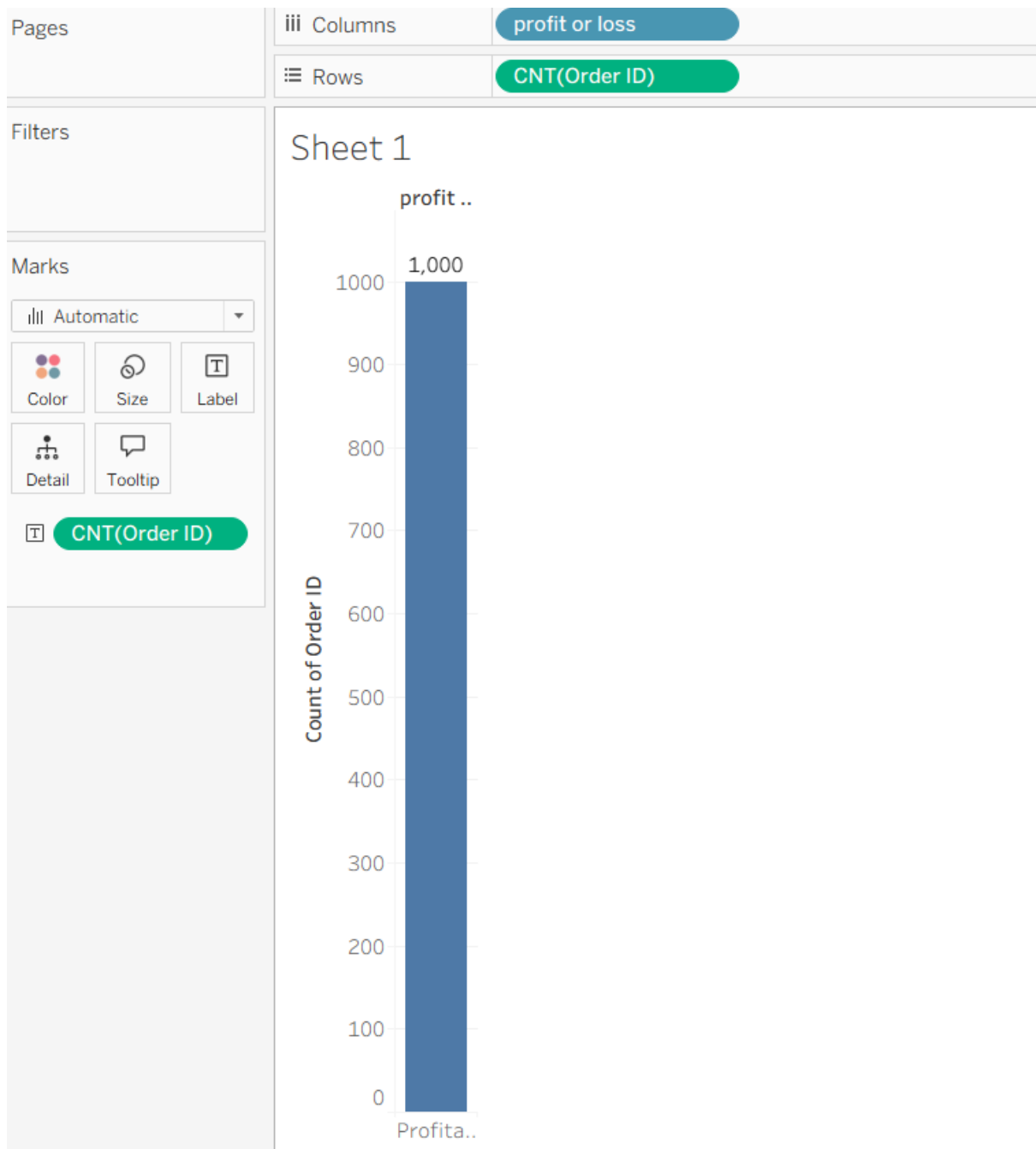
IF [Profit] < 0 THEN "Loss-Making"
ELSE "Profitable"
END

```

The calculation is valid.

#	47	Card	#	800	568.971	231.030	delayed	Profitable
Sheet1	17	Sheet1	Sheet1	212	Sheet1	120	Calculation	Calculation
Delivery Time Min	17	Payment Mode	Sales	Cost	Profit	order classification	profit or loss	
15	UPI		276	228.359	47.641	fast	Profitable	
23	Cash		196	145.479	50.521	fast	Profitable	
41	Card		240	189.731	50.270	delayed	Profitable	
44	Card		220	176.649	43.351	delayed	Profitable	
19	UPI		524	359.182	164.818	fast	Profitable	
12	UPI		348	246.531	101.469	fast	Profitable	

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Task 3: Top N Products Analysis (Parameter)

Business Need

The product team wants to focus on top-performing products.

Task

Create a Top N Products dynamic analysis.

Answer :

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Edit Parameter [top n product] ×

Name

top n product

Properties

Data type

Integer

Display format

5

Current value

5

Value when workbook opens

Current value

Allowable values

☐ All ☐ List ☒ Range

Range of values

☒ Minimum 1

☒ Maximum 50

☒ Step size 1

☒ Fixed

☐ When workbook opens

Add values from ▼

Cancel

OK

rank ×

Results are computed along Table (across).

`RANK(SUM([Sales])) <= [top n product]`

Default Table Calculation

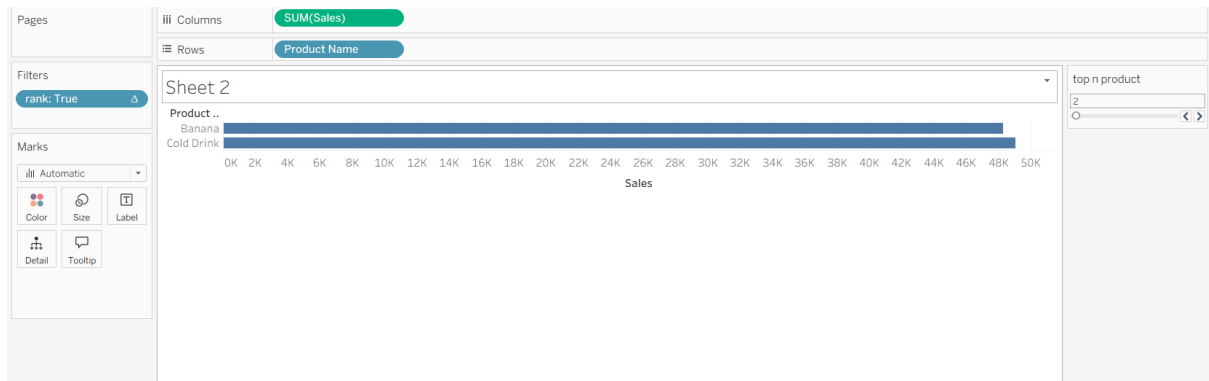
The calculation is valid.

1 Dependency ▼

Apply

OK

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Task 4: City-Level Benchmarking (FIXED LOD)

Business Need

Compare city performance independent of category filters.

Task

Calculate average sales per city using FIXED LOD.

Answer :

```
{ FIXED [City] : AVG([Sales]) }
```

The calculation is valid.

Apply

OK

Task 5: Customer/Product Detail Impact (INCLUDE LOD)

Business Need

The sales team wants average sales considering product-level detail, even when product is not shown.

Task

Create a calculation that finds the average sales at the product level, but allows the result to be used at a higher level (such as Category or Region). Use INCLUDE LOD.

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Answer :

×

```
{ INCLUDE [Product Name] : AVG([Sales]) }
```

The calculation is valid.

Apply

OK

Double-click or drag fields from th